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Education

M.MATH. Student in Applied Mathematics (Fluid Mechanics Group). University of Waterloo, Waterloo, Ontario, Canada. Ranked: [#112 in the world overall, #43 in the world in Mathematics, and #37 in the world in Engineering, all according to QS Global World Rankings](#). (Started in Fall 2023)

CGPA 81.50 (A) out of 100

M.Sc. in Mechanical Engineering Energy Conversion. Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran. Ranked: [#3rd best university in Iran according to the QS Global World Ranking](#), and [#32 in the world in Mechanical Engineering according to the U.S. News & WORLD REPORT Subject Rankings](#).

CGPA 17.65 (A) out of 20 on the Iranian scale or **3.9** on the American scale [according to WES](#).

Thesis Title Experimental and Numerical Study of Deformation of Fibers in a Viscous Fluid. (Sep. 2022)

B.Sc. in Mechanical Engineering, Kermanshah University of Technology, Kermanshah, Iran.

CGPA 14.04 (B) out of 20

Thesis Title Calculating Stress Intensity Factor In Symmetric Crack of Perforated Plate With Inside Pressure. (Sep. 2018)

Internship

- Quality Engineering Co-op at [Westport Fuel Systems](#) Cambridge, ON, Canada (September 2025 — April 2026)
- Research assistant at the Hydrodynamics Lab (LadHyX), Ecole Polytechnique, Institut Polytechnique de Paris [ranked #38 in the world in QS Global World Ranking](#), Paris, France. (July 2021 — October 2021)

List of Publications

- **Sedimenting Flexible Fibers Against Obstacles in a Viscous Fluid**

U Makanga, M Sepahi, C Duprat, B Delmotte

[*Bulletin of the American Physical Society*](#)

- **Obstacle-induced lateral dispersion and nontrivial trapping of flexible fibers settling in a viscous fluid**

Ursy Makanga, Mohammadreza Sepahi, Camille Duprat, and Blaise Delmotte

Physical Review Fluids 8 (4) 044303 – Published 20 April 2023

DOI: <https://doi.org/10.1103/PhysRevFluids.8.044303>

- **Sedimentation of Flexible Fibers in Structured Environments**

Ursy Makanga, Mohammadreza Sepahi, Camille Duprat, and Blaise Delmotte

SIAM Conference on Computational Science and Engineering –Published February 27, 2323

<https://hal.science/hal-04438332>

- **Sedimenting flexible fibers against obstacles in a viscous fluid**

C Duprat, U Makanga, M Sepahi, B Delmotte

APS DFD meeting 2022

<https://hal.science/hal-04574057/>

Languages

TOEFL iBT: 106 (Reading 28, Listening 30, Speaking 26, Writing 22), April 12, 2023

Persian: Native

Research Interests

- Fluid Dynamics (ML, Numerical and Experimental)
- Wind Energy, Offshore Wind Turbines
- Physical Oceanography, Atmospheric Science, and Climate Dynamics
- Biological Fluid Dynamics

Awards and Honors

- Head TA position at the University of Waterloo, Waterloo, ON, Canada.
- Graduate Research Studentship (GRS), University of Waterloo, ON, Canada.
- Waterloo's President's Graduate Scholarship, University of Waterloo, ON, Canada.
- International Doctoral Student Award (IDSA) \$15,420, University of Waterloo, ON, Canada.

- Earned the Erasmus+ KA107 fellowship for the Hydrodynamics Lab (LadHyX) at Ecole Polytechnique, Institut Polytechnique de Paris, Paris, France (Summer 2021).
- Ranked in the top 10 percent for CGPA amongst the graduating class (35 students) started in 2019 in the Master's program at Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran.
- Studied both bachelor's and master's degrees tuition-free
- Winning 1st place in the seventh round of Iranian students' debate competitions (2018).

Teaching Experience

- TA, Running Calculus tutorials (Spring 2024, Fall 2024, Spring 2025, Spring 2026)
- Head TA Calculus 2, University of Waterloo (Winter 2024)
- TA, Theoretical Mechanics, University of Waterloo (Fall 2023).
- TA, Introduction to Differential Equations, University of Waterloo (Fall 2023).

Computer Skills

- MATLAB, Python, COMSOL Multiphysics, Linux Shell, ImageJ, WRF, ML, LaTeX, OpenFOAM

Selected Courses

Graduate

- Advanced Mathematics, GPA: 18/20 (A+)
- Advanced Fluid Mechanics, GPA: 17/20 (A)
- Hydrodynamic Instability, GPA: 19/20 (A+)
- Turbulence, GPA: 19/20 (A+)
- Mathematics for Climate Change, GPA: 91/100 (A+)
- Scientific Machine Learning, GPA: 92/100 (A+)

Hobbies

- Gym, Soccer, Cycling, Swimming, Camping,

Certificates

- Winter school on “Energy Systems, Resilience, and Climate Change” at the Sharif University of Technology, Tehran, Iran, (February 2021).
- Online Summer School on “Linux Shell” presented by Campute Ontario (June 2024)